Computational Modeling

BMED365: Topic List Z01–Z13

BMED365

Department of Biomedicine, University of Bergen

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Overview

Z01: Explain the role of mathematical models in biomedicine

Why Mathematical Models in Biomedicine?

Definition

A **mathematical model** is a description of a system using mathematical concepts and language – enabling simulation, prediction, and understanding of biological processes.

Key Roles:

- **Understanding:** Reveal mechanisms
- **Prediction:** Forecast outcomes
- **Optimization:** Design interventions
- **Integration:** Combine data types
- **Hypothesis testing:** Virtual experiments

Applications:

- Drug pharmacokinetics
- Disease progression
- Cardiac electrophysiology
- Tumor growth dynamics
- Neural activity
- Epidemiology

The Digital Twin Vision

Z02: Distinguish between mechanistic and phenomenological models

Two Fundamental Modeling Approaches

Mechanistic Models:

- Based on physical/biological principles
- Parameters have biological meaning
- Derived from first principles
- More complex, harder to fit
- Better extrapolation

Example:
Hodgkin-Huxley model – ion channels, conductances, Nernst potentials

Phenomenological Models:

- Describe observed behavior
- Parameters may lack meaning
- Fitted to data empirically
- Simpler, easier to use
- Limited extrapolation

Example:
Logistic growth – $\frac{dN}{dt} = rN(1 - \frac{N}{K})$

Practical Choice

Use **mechanistic** models when you need to understand *why*; use **phenomenological** models when you need to predict *what*.

Z03: Describe ordinary differential equations (ODEs) in biological systems

ODEs: The Language of Dynamic Systems

What is an ODE?

An equation relating a function to its derivatives:

$$\frac{dy}{dt} = f(y, t, \theta)$$

where y is the state, t is time, and θ are parameters.

Common in biology:

- Population dynamics
- Chemical kinetics
- Drug metabolism
- Gene regulation
- Neural firing

Solving ODEs:

- Analytical (rare in biology)
- Numerical integration
- Euler, Runge-Kutta methods
- Python: `scipy.integrate`

Z04: Explain the Hodgkin-Huxley model of action potentials

The Nobel Prize-Winning Model (1963)

Core Idea

The action potential arises from voltage-dependent opening and closing of ion channels (Na^+, K^+) in the cell membrane.

The HH Equations:

$$C_m \frac{dV}{dt} = I - g_{Na} m^3 h(V - E_{Na}) - g_K n^4 (V - E_K) - g_L (V - E_L)$$

Components:

- V: membrane potential
- m, h, n: gating variables
- g: conductances
- E: reversal potentials

Action Potential Phases:

1. **Resting:** $V \approx -70$ mV
2. **Depolarization:** Na^+ influx
3. **Repolarization:** K^+ efflux
4. **Hyperpolarization:** Overshoot
5. **Recovery:** Return to rest

Lab 5

Notebook 01: Simulate HH model and explore parameter effects.

Modeling Blood Flow and Cardiac Signals

Z07: Cardiovascular Flow Models

Windkessel Model:

- Arteries as elastic reservoir
- Resistance + Compliance
- Predicts pressure waveforms

1D Blood Flow:

- Navier-Stokes simplification
- Wave propagation in arteries
- Pulse wave velocity

Z08: ECG and Blood Pressure

ECG Modeling:

- Cardiac dipole theory
- P, QRS, T waves
- Lead configurations

Pressure Waveforms:

- Systolic/diastolic peaks
- Dicrotic notch
- Arterial stiffness markers

Lab 5 Notebook 03

Simulate cardiovascular models with ECG and blood pressure waveforms.

Z05–Z06: Tumor growth and angiogenesis models

Mathematical Oncology

Z05: Tumor Growth Models

Classic Models:

- **Exponential:** $\frac{dN}{dt} = \lambda N$
- **Logistic:** $\frac{dN}{dt} = \lambda N(1 - \frac{N}{K})$
- **Gompertz:** $\frac{dN}{dt} = \lambda N \ln(\frac{K}{N})$

Parameters:

- λ: growth rate
- K: carrying capacity
- N: tumor size/cell count

Z06: Angiogenesis Models

Why it matters:

- Tumors need blood supply
- Anti-angiogenic therapy
- Metastasis pathway

Modeling approaches:

- VEGF diffusion
- Endothelial cell migration
- Vessel network formation
- Nutrient delivery

Clinical Relevance

Tumor growth models inform treatment scheduling, dose optimization, and predict response to therapy

Z09–Z10: Muscle force and cell signaling

Biomechanics and Systems Biology

Z09: Muscle Force Models

Hill Model:

- Force-velocity relationship
- Contractile + elastic elements
- Activation dynamics

Applications:

- Movement biomechanics
- Prosthetics design
- Sports performance
- Cardiac muscle function

Z10: Cell Signaling Networks

ODE-based modeling:

- Receptor-ligand binding
- Enzyme kinetics (Michaelis-Menten)
- Phosphorylation cascades
- Feedback loops

Gene Regulatory Networks:

- Transcription factor dynamics
- Oscillations (circadian)
- Bistability (cell fate)

Lab 5 Notebooks

04: Muscle force models 05: Cell signaling and gene regulation

Z11: Describe how to use LLMs to assist model development

LLMs as Modeling Assistants

How LLMs can help:

- Generate model equations
- Write simulation code
- Explain model behavior
- Debug numerical issues
- Literature review
- Parameter estimation ideas

Example prompt:

"Write Python code to simulate the Hodgkin-Huxley model using scipy.integrate.solve_ivp"

Best practices:

- Verify generated equations
- Check units and dimensions
- Test with known solutions
- Iterate on prompts
- Combine with domain expertise

Lab 5 Notebook 06

Use LLM to generate a kidney filtration model from scratch – demonstrating AI-assisted modeling workflow.

Z12–Z13: Systems biology and model validation

Advanced Modeling Concepts

Z12: Systems Biology Approaches

Multi-scale modeling:

- Molecular → Cellular
- Cellular → Tissue
- Tissue → Organ
- Organ → Organism

Integration challenges:

- Different time scales
- Parameter transfer
- Computational cost

Z13: Model Validation

Validation approaches:

- Compare with experimental data
- Sensitivity analysis
- Uncertainty quantification
- Cross-validation

Key questions:

- Does the model fit the data?
- Are predictions robust?
- What are the limitations?

Resources

BioModels: Database of curated models – https://www.ebi.ac.uk/biomodels/

Summary: Computational Modeling

Key Learning Points:

- **Z01–Z03:** Fundamentals – mechanistic vs. phenomenological, ODEs
- **Z04, Z07–Z08:** Electrophysiology – Hodgkin-Huxley, cardiovascular
- **Z05–Z06:** Oncology – tumor growth, angiogenesis
- **Z09–Z10:** Biomechanics – muscle, cell signaling
- **Z11–Z13:** Advanced – LLM-assisted, systems biology, validation

Lab 5 Notebooks

00-test-llm	Local LLM setup (DeepSeek-R1)
01-action-potentials	Hodgkin-Huxley model
02-tumor-growth	Tumor dynamics
03-cardiovascular-flow	Blood flow + ECG
04-muscle-force	Force generation
05-cell-signaling	Gene networks
06-kidney-filtration	LLM-assisted modeling

Individual Project

Consider modeling projects: cardiac, neural, tumor, pharmacokinetics